

LINE GO: Maximizing the convenience of ride-hailing services with Google Maps Platform

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ABOUT LINE GO

LINE GO

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By providing accurate geospatial data and comprehensive navigation features, LINE GO Google Maps Platform enhance the convenience of its ride-hailing services and drivers' work efficiency.

About LINE GO

LINE GO is the only ride-hailing company in Taiwan offering services through LINE, the country's most popular messaging app. Founded in 2017, it continues expanding its business range to provide maximal mobility convenience.

Industries: Transportation & Logistics,
Travel & Hospitality

Location: Taiwan

Google Maps Platform results

- Supports quick location search with the Places API
- Provides precise precalculated fares to promote digital payment through the Distance Matrix API
- Boosts driver dispatch efficiency with highly accurate ETAs

Enhanced
accuracy
20%



About Master Concept

Master Concept is a Google Maps Platform and Google Cloud partner providing cloud strategy, implementation, and integration support to enterprises across all industries in APAC.

Google Maps Platform

(<https://mapsplatform.google.com/>)

Directions API

(<https://developers.google.com/maps/documentation/directions/start>)

Distance Matrix

(<https://developers.google.com/maps/documentation/distance-matrix/start>)

Google Places API

(<https://mapsplatform.google.com/maps-products/>)

Ride-hailing applications are loved in Taiwan for their convenience, but the riding experiences they offer can be compromised due to a lack of localized features. At the same time, taxi drivers are constantly looking for on-demand cab platforms that allow them to gain higher revenues and work

more efficiently. LINE GO (<https://linetaxi.com.tw/>) sets itself apart from other ride-hailing service providers in Taiwan by addressing these priorities for local passengers and drivers

Founded in 2017, it has been the only company in Taiwan providing ride-hailing service on LINE, the most popular

(<https://www.statista.com/statistics/968238/taiwan-communication-software-usage/#:~:text=Popular%20instant%20messaging%20platforms%20in%20Taiwan%202023&text=LINE's%20dominance%20in%20Taiwan's%20instant,market%20share%20of%203.6%20percent.>)

messaging app in the country. By communicating with LINE GO's chatbot, LINE users can swiftly complete the ride-hailing process, from inputting locations to payment, with just a few clicks. On the drivers' end, LINE GO offers lower partnership fees and enables shorter idle time to help drive up revenues. All these efforts have paved the way for rapid growth. Within six years, the startup has accumulated 13,000 partner drivers and 3.5 million member users across Taiwan.



"We're dedicated to improving the cab industry through innovative use of technology," says Kevin Chan, LINE GO's CEO. "That's why we're always finding new ways to provide the most convenient services to passengers and the best work conditions to drivers." With this in mind, LINE GO sought the most advanced geolocation technology when building its service. The startup decided to employ Google Maps Platform

(<https://mapsplatform.google.com/>), because of its comprehensive set of geolocation tools supported by frequently updated data.

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Founder & CTO, LINE GO

"Our team had long been using the Google Maps consumer app for daily navigation, so we already had high confidence in the accuracy level of the geolocation data on Google Maps Platform before adopting it," explains Hayden Huang, LINE GO's CTO. "Besides, compared with other maps services, it provides more comprehensive geolocation tools that can support our continual development of new features."

Enabling quick and easy ride planning with highly accurate geolocation data

To help customers plan their rides more smoothly, LINE GO needs to provide real-time ETAs that are as accurate as possible. At the development stage, the team used various geolocation tools to calculate

ETAs and then compared the results with the real time of arrival. The test showed that the Distance Matrix API

(<https://developers.google.com/maps/documentation/distance-matrix/overview>)

delivers ETAs that are 20% more accurate than other offerings, since it takes into account constantly changing factors like traffic and weather conditions. Leveraging the Distance Matrix API, LINE GO is able to build customer trust by minimizing unexpected delays for passengers traveling during rush hours in big cities.

"The accuracy of ETAs is crucial to our service quality, because that's what our customers rely on to ensure that they can get to their destinations on time," notes Huang. "The highly accurate ETAs provided by Google Maps Platform based on real-time factors have allowed us to provide highly reliable services."



Hayden Huang, Co-Founder & CTO, LINE GO

Another way LINE GO facilitates ride planning is by supporting quick location search through the Places API

(<https://developers.google.com/maps/documentation/places/web-service/overview>)

. When users start inputting a pickup or dropoff location in LINE GO's service, the autocomplete feature enabled by the Places API instantly generates the most relevant location suggestions, which helps reduce the time required to place a ride order. On top of that, according to LINE GO's comparison test, the geolocation data on Google Maps Platform is updated more frequently. Therefore, its customers can easily find places, like newly opened restaurants, in its service.

Enhancing drivers' trust in precalculated fares with the Distance Matrix API

Since the release of its ride-hailing app, LINE GO has been improving its services by expanding and optimizing its service features. At first, LINE GO's users and drivers could only rely on in-taxi meters to calculate fares, which made it more challenging to adopt in-app payment and create a smooth user experience. In 2019, the startup began to offer an option to pay the fare that is calculated before a ride, based on the most optimal route generated by the Distance Matrix API. When customers choose this option, the planned route is shown to drivers through the Directions API

(<https://developers.google.com/maps/documentation/directions/overview>)

, allowing them to follow the path suggested by the Distance Matrix API and get paid correctly.



*Hayden Huang, Co-Founder & CTO, LINE GO and Kevin Chan,
Co-Founder & CEO, LINE GO*

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Huang notes that the route planning precision plays a pivotal role in the successful launch of the feature, because the precalculated fares need to be accurate enough to win drivers' trust. With the Distance Matrix API, LINE GO is able to convince its drivers that the fare calculation model is trustworthy and can promote digital payment. Before the adoption of the Distance Matrix API for fare calculation, the startup used to receive frequent complaints from drivers that the precalculated rates were inaccurate. Now, the number of such reports has decreased by 30%.

"We understand that taxi drivers want to ensure the route suggested to them in our service is the most ideal one at the time, and they're not underpaid due

to factors like unexpected traffic jams," Huang adds. "The Distance Matrix API provides a highly reliable route planning feature to accurately calculate ride rates in advance. This way, we're able to offer a more streamlined ride-hailing process, because passengers can directly pay in our service, instead of waiting for in-taxi meters to show the price at the end of a ride."

Increasing driver dispatch and operational efficiency

When a passenger places a ride order through LINE GO's platform, the team assigns the order to the driver who's nearest to the pickup location. It used to do so based on straight-line distance, which doesn't account for the real driving time required when viaducts or bridges are on the route. As a result, LINE GO's customer service often received calls from drivers saying that they were not able to pick up their customers as soon as expected.

To dispatch its drivers more efficiently, LINE GO, in early 2023, began to rely on the ETAs generated by the Distance Matrix API to assign orders, resolving 90% of dispatch issues caused by imprecise ETA calculations for routes involving viaducts or bridges. With the new feature, LINE GO can now more correctly discern the driver who can arrive first by accurately identifying journey distances and which drivers can arrive at the pickup location in the shortest time.



The LINE GO team

"We're dedicated to helping drivers maximize their revenues, so it's important that our system provides the most accurate geolocation data. Using the Distance Matrix API to find the driver that can complete an order in the fastest manner, we've helped them work more efficiently and enhanced our overall operational efficiency," says Huang.

"With its highly
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—*Kevin Chan* , Co-
Founder & CEO, LINE GO

Expanding features for greater mobility convenience

On a constant quest to improve its ride-hailing services, LINE GO has been working closely with Master Concept (<https://hkmci.com/>), a Google Maps Platform partner. When LINE GO needed to overcome an obstacle or develop a new feature, the Master Concept team was always there to offer the most suitable Google Maps Platform solution for the use case, such as leveraging the Distance Matrix API to increase driver dispatch efficiency.

Next, LINE GO plans to employ On-demand Rides and Deliveries solution

(<https://developers.google.com/maps/documentation/transportation-logistics/on-demand-rides-deliveries-solution>)

to provide drivers with in-app navigation that follows the best route suggested by the Distance Matrix API. This way, its drivers will no longer have to juggle between the LINE GO and the Google Maps consumer apps during a ride, which can help save time and enhance driving safety.

At the same time, the startup continues enhancing the riding experience that it offers while promoting sustainability. In October 2022, LINE GO expanded its electric fleet and launched a premium ride option with the latest electric car model, internally trained drivers and advanced in-taxi services, such as phone charging.

Chan concludes, "With its highly accurate geolocation data and extensive features, Google Maps Platform empowers us to offer excellent user experience to both passengers and drivers through continuous service improvement. We're now set to bring our services to the next level and enable higher mobility convenience."